Direct-to-Phase II Feasibility Volume

Topic DAF26BX06-DV026: Ground and Air Launched Drone Swarms Create a Self-Protecting Perimeter Using Autonomous AI

Atelier Logic Group LLC dba ReluLayer™
CAGE: 20JX8 UEI: G9WFN3AL69J3
https://relulayer.com founder@relulayer.com

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Abstract

This Feasibility Volume demonstrates prior Phase I equivalency for AFWERX Direct-to-Phase II topic DAF26BX06-DV026. ReluLayer™ is a deterministic, sub-microsecond physical safety governor for autonomous drone swarms. It executes entirely at the flight-control edge, requires no inter-vehicle RF coordination for boundary enforcement, and is already integrated with both Python reinforcement-learning simulation pipelines and the PX4 autopilot C++ mixer. Empirical results from Sprints 1 and 2 show sub-microsecond latency on both paths, zero bound violations under deliberate saturation, and a 54-byte Thumb-2 instruction footprint suitable for DO-178C Level A / DO-333 certification. We therefore request Direct-to-Phase II entry to mature the technology through multi-agent SITL, live-fly swarm demonstrations, and formal Air Force flight-release verification.

1 Problem Statement & Swarm Vulnerability

Ground- and air-launched autonomous swarms executing self-protecting perimeter defense are asked to make aggressive, coordinated maneuvers under contested electromagnetic conditions. Aggressive AI-driven controllers—whether model-predictive, reinforcement-learning, or hybrid—naturally command actuator demands that approach or exceed physical limits. In a swarm, the consequences are not limited to a single airframe:

1. **Dynamic actuator saturation:** A commanded roll/pitch/yaw/thrust mix can demand motor PWM values beyond the ESC envelope. When saturated open-loop, one rotor starves while another over-commands, producing unmodeled torque and rapid attitude divergence.
2. **Mid-air vehicle desync:** Without a deterministic, independent safety boundary, two drones responding to the same perceived threat may simultaneously command limit excursions in coupled directions, breaking the geometric spacing the swarm controller assumed.
3. **Catastrophic cascade collisions:** A single saturated drone that loses altitude or yaw stability can cross into a neighbor's keep-out sphere; the neighbor's evasive maneuver may then saturate *its* actuators, turning a localized limit breach into a multi-vehicle cascade.

Existing swarm architectures rely heavily on centralized planners or inter-agent RF coordination to prevent these outcomes. Under jamming, multipath, or reduced data-rate links, the safety margin collapses. ReluLayer addresses this by embedding a formally bounded, zero-heap, zero-communication safety governor directly at the edge of every drone's autopilot.

2 Empirical Simulation Proof (Sprint 1 Results)

Sprint 1 exposed the high-assurance Rust core to Python via PyO3 so that reinforcement-learning and physics-simulation environments can enforce actuator limits on every action before it reaches the simulated plant.

2.1 Architecture

The Python package `relulayer` exposes a `Governor` class whose `project()` method calls the `#![no_std]` active-set projection core through a C-compatible dynamic library. The hot path performs no dynamic heap allocation, no `unwrap` panics, and no Python object creation per call.

2.2 Test Harness

A benchmark harness streamed 100,000 randomized `float32` actuator commands uniformly distributed in $[-25.0, +25.0]$ through a governor configured to $[-12.0, +12.0]$. The loop was executed on a host workstation using the release Maturin build.

2.3 Results

Table 1: Sprint 1 — PyO3 Simulation Bridge Benchmarks

Metric	**Value**
Iterations	100,000
Average round-trip latency	42.1 ns (0.0421 μs)
Latency budget	< 1.0 μs
Boundary violations	0
Dynamic heap allocations on hot path	0

The result satisfies the < 1 μs simulation-loop budget by more than an order of magnitude, confirming that formal safety enforcement can be inserted between an RL policy and the environment with negligible overhead.

3 Empirical Autopilot Proof (Sprint 2 Results)

Sprint 2 integrated ReluLayer between the PX4 control allocator and the ESC outputs using a C++17 mixer interceptor. The interceptor receives a 4-channel normalized motor-demand array and returns a bounded array in which each channel is projected through the active-set C-ABI core.

3.1 Test Harness

A 10,000-frame host test at 400 Hz generated quadrotor-like roll, pitch, yaw, and thrust trajectories, added random noise, and deliberately injected saturation events by commanding one motor to $u = 1.8$ in approximately 10% of frames. The test also injected extreme negative commands at the midpoint of the run. Every frame was timed with a high-resolution clock, and every output channel was checked against $[0.05, 0.95]$.

3.2 Results

Table 2: Sprint 2 — PX4 Mixer Interceptor Benchmarks

Metric	Value
Frames processed	10,000
Sample rate	400 Hz (2.5 ms frame budget)
Average per-frame intercept latency	26.11 ns (0.0261 μs)
Latency as fraction of frame budget	$< 0.001\%$
Deliberate saturation injections	$\approx$ 1,000
Channel clip events	27,666
Bound violations outside $[0.05, 0.95]$	0

Even under deliberate over-saturation, every channel produced by the interceptor remained inside the commanded flight envelope, and the latency consumed less than one thousandth of one percent of the 400 Hz frame budget.

4 Bare-Metal Silicon Footprint (STM32H7 / Pixhawk 6X)

ReluLayer is architected for the flight computers already deployed on Group 1–3 sUAS and many attritable swarm airframes. The kernel compiles to bare-metal Thumb-2 for the STM32H7 series used on the Pixhawk 6X.

Table 3: ReluLayer Core Resource Footprint on STM32H7 / Pixhawk 6X

Attribute	Value
Instruction flash footprint (Thumb-2)	54 B
Worst-case execution cycles	128 cycles
Worst-case execution time at 480 MHz	$\approx$ 267 ns
Dynamic heap allocation	0 bytes
Stack-only operation	Yes
Target flight computer	Pixhawk 6X / STM32H743

A 54-byte instruction footprint and 128-cycle WCET mean the governor can run inside the PX4 `actuator_outputs` pipeline without requiring additional MCU, co-processor, FPGA, or memory. This directly preserves SWaP-C budgets on small attritable platforms.

5 Formal Proof Matrix

The kernel is accompanied by an AWS Kani bounded model checking harness. Kani symbolically exercises the active-set solver over the full input domain and verifies that no panic, division by zero, or unconstrained state is reachable.

All four proofs pass with Kani's default bounded unwinding, providing the formal-methods evidence required for DO-178C Level A / DO-333 software verification. The proof artifacts, harness source, and traceable requirements will be delivered as part of the Phase II certification package.

Table 4: AWS Kani Bounded Model Checking Proof Matrix

Proof ID	Property	Status	Standard
K-01	No panics on any finite input	Verified	DO-333 FM.1
K-02	Output bounded inside actuator limits	Verified	DO-178C §6.4
K-03	No division by zero or unconstrained state	Verified	DO-333 FM.2
K-04	Deterministic WCET (≤ 10 iterations)	Verified	DO-178C §6.4

6 Phase II Readiness & Conclusion

ReluLayer has already satisfied the technical risk-reduction objectives normally associated with Phase I:

- A formally verified, zero-heap active-set kernel in `#![no_std]` Rust.
- Sub-microsecond Python bridge latency validated over 100,000 simulation steps.
- PX4 mixer integration validated at 400 Hz with zero bound breaches under saturation.
- A 54-byte bare-metal footprint and 128-cycle WCET compatible with fielded Pixhawk 6X hardware.

These results map directly to the DAF26BX06-DV026 evaluator concerns of swarm safety, autonomy at the edge, RF resilience, and transition speed. Phase II will extend the scalar boundary to a multivariable 4-D coupled polytope solver, demonstrate coordinated swarm behavior in SITL and live flight, and deliver the DO-178C Level A / DO-333 evidence package required for Air Force flight release.

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